

# LTE: A Natural Choice for Long-term IoT or M2M Communication

Many machine-to-machine (M2M) operators consider LTE as their natural choice for long-term Internet of Things (IoT) or, to be precise, for M2M communications

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The 'things' in the IoT, or the 'machines' in the M2M, are physical entities whose identity or state (or the state of whose surroundings) is capable of being relayed to an Internet-connected infrastructure. M2M refers to the machine-to-machine devices which span over a wide variety of telematics (where the host device is portable or mobile) and telemetry (where the device is stationary) applications and services.

## LTE and M2M applications

Long-term evolution (LTE) is the fastest growing standard in the history of telecom industry, and the latest-generation mobile network technology. The LTE advanced standard is an enhancement to the LTE and is also known as 4G LTE.

Although the typical data rates for M2M communication are lower than the data rates for which LTE has been designed, some specific M2M applications can be created or enhanced to take advantage of the available bandwidth.

## Key features of LTE

LTE offers the following features over 2G and 3G networks for better communication between the devices:

**High gross bandwidth.** It provides a high bandwidth of up to 150 Mbps down and 50 Mbps up in today's net-

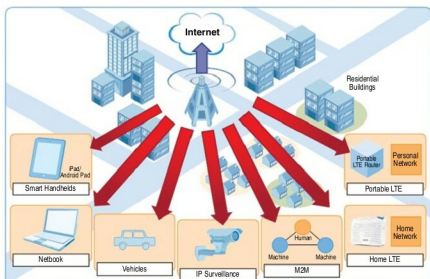


Fig. 1: Various applications in the M2M market that benefit from LTE (Courtesy: <http://www.zyxel.com>)

works, which is sufficient to stream eight simultaneous HDTV channels at a time.

**Efficient sharing.** LTE allows efficient sharing of the gross bandwidth between large number of users, allowing cost-effective allocation of data to different requirements of devices. It enables simultaneous handling of low-bandwidth traffic.

**Reduced operational expenditure (OPEX).** LTE reduces the operational expenditure through a simplified network architecture and better spectral efficiency. Thus there is a high rate of information transfer over a given bandwidth at low expenditure.

**Low latency.** It guarantees a maxi-

mum latency of around 10 milliseconds, below the human perception, which supports a faster communication.

## Benefits of LTE in M2M market

**Longevity.** The lifecycle of older cellular networks is unpredictable and the cost of upgrading can be extraordinarily high. LTE will support connected devices in the field for many years to come.

**Potentially lowered service costs.** LTE is much more spectrum efficient and less expensive to operate than the previous-generation technologies.

**Scalability.** Most LTE networks deployed today support IPv6, making

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